Pole Locations 1

EGR 345

GVSU

Pole Locations Intro Questions

- what is a pole location?
- why are pole locations so important in system dynamics and feedback control?
- how do ζ and ω_n relate to second-order pole locations?
- how do ζ and ω_n affect the step response of a system?
- how do I quantify the step response of a system?

What is a pole?

What are the poles of a second-order, under-damped TF?

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

What does it look when we plot second-order poles?

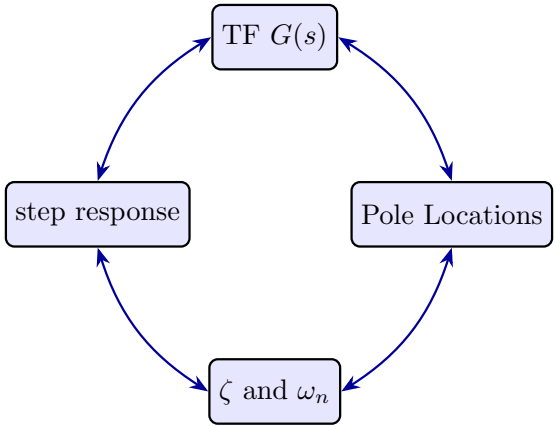
$$s = -\zeta\omega_n \pm j\omega_d$$

Learning Goals for Pole Locations

- given a sketch of pole locations for second-order TFs:
  - estimate ζ and ω_n
  - roughly sketch the step response in your head or “by hand”
- given ζ and ω_n:
  - plot the pole locations on a real vs. imag. axis
  - sketch the step response
- given the step response of a TF:
  - estimate ζ and ω_n
  - sketch the pole locations

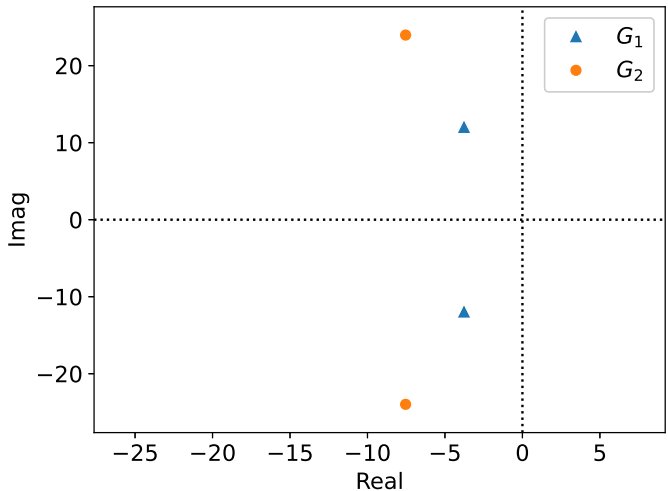
Pole Locations Cycle

- given any *one* of these, we can find *all* the others

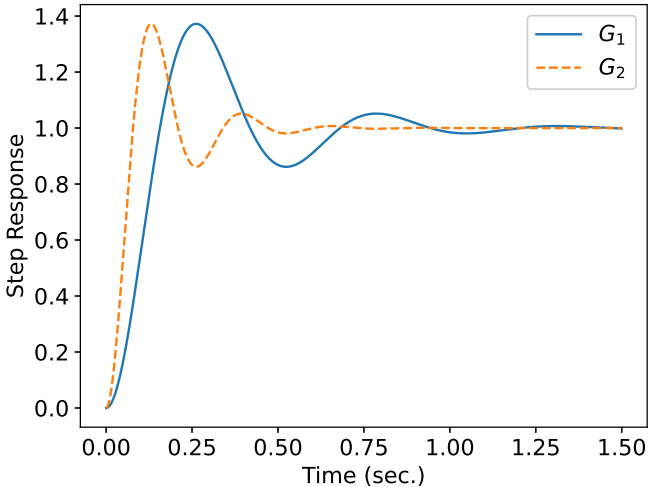


Estimating Step Response from Pole Locations

- how do the step responses compare for G_1 and G_2?

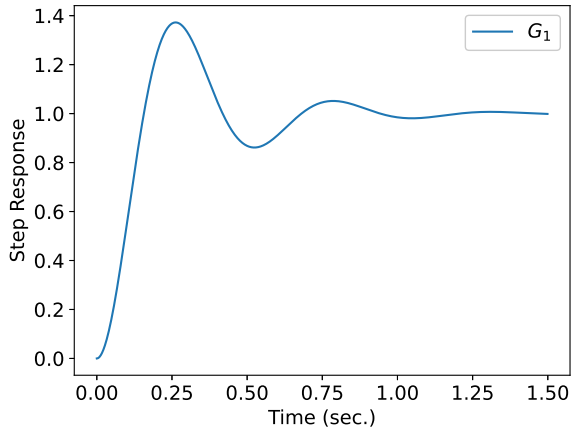


Answer Check for Step Responses of G_1 and G_2



Quantifying a Step Response

- how do we quantify a “typical” second order step response?



How do ζ and ω_n connect to our step response measures?

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- Find $y_{step}(t)$